

## ***Malicious URL/Phishing Detector***

### **Problem Description**

Phishing is a dangerous scam that is used to steal people's information. With the continuous growth in Phishing attacks across different use cases (email, suspicious links, etc.) it is becoming important to detect phishing quickly in order to scale with the rate it is being used. We seek to determine if Bloom Filtering is an efficient and scalable way to deal with the problem.

### **Data Source**

<https://www.kaggle.com/datasets/aksingh2411/dataset-of-malicious-and-benign-webpages/code>

We will filter this dataset to only include malicious webpages, which will later be used for membership testing.

### **Algorithm**

Bloom Filter, a probabilistic data structure that can quickly check membership. It can be used on a dataset of malicious links to check if a new link is in that dataset, meaning that it would be malicious. Since it has no false negatives, it will catch all known malicious links.

### **Evaluation Methods**

- Comparison of Time and memory performance of using bloom filter for membership checking vs not using the bloom filter
- Percentage of Accurate Predictions while using Bloom Filter

### **Expected Outcomes**

We will know whether a bloom filter by itself is strong enough to prevent phishing, or if other algorithms are needed in addition to it.